
IE7860: Advanced Analytics Methods

Predicting GDP Cycle Direction

Using Macroeconomic Indicators and Financial News Sentiment

A Walk-Forward Gradient Boosting Approach with FinBERT Sentiment

Course IE7860: Advanced Analytics Methods

Group Group 4

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Members Bao Phan, Jacob White, Tianxiang Jiang, Suhad Alnoor, Malak Amireh

Executive Summary

This report presents a machine learning framework for predicting the direction of the U.S. GDP business cycle on a quarter-ahead horizon. Using 35 years of quarterly macroeconomic data (1990 to 2025) from the Federal Reserve Economic Data (FRED) database and Yahoo Finance, we decompose log real GDP into trend and cycle components via the Hodrick-Prescott (HP) filter and an Unobserved Components (UC) state-space model.

The forecasting task is framed as binary classification: will the UC cycle rise or fall next quarter? A Gradient Boosting Classifier evaluated under a strict walk-forward expanding-window protocol achieves 54.3% accuracy and 0.431 F1 score, outperforming a naive persistence baseline (43.3% accuracy). The sentiment pipeline, pulling headlines via Alpha Vantage and scoring with FinBERT, was fully implemented; however, the free-tier API provides only recent articles, leaving insufficient overlapping quarters to conduct a valid walk-forward comparison against the macro-only model.

Key contributions: (1) a rigorous HP vs. UC cycle comparison, (2) a walk-forward evaluation framework that eliminates lookahead bias, (3) real-time news sentiment integration using transformer-based NLP, and (4) interpretable feature importance analysis.

Table of Contents

Executive Summary	2
Table of Contents	3
1. Introduction	4
1.1 Research Questions	4
1.2 Scope and Limitations	4
2. Data Sources and Assembly	5
2.1 Macroeconomic Series (FRED)	5
2.2 Market Series (Yahoo Finance)	5
2.3 Data Quality	5
3. Trend-Cycle Decomposition	6
3.1 Hodrick-Prescott (HP) Filter	6
3.2 Unobserved Components (UC) Model	6
3.3 HP vs. UC Comparison	6
4. Sentiment Analysis Framework	7
4.1 Data Collection	7
4.2 FinBERT Scoring	7
4.3 Sentiment Visualizations	8
5. Feature Engineering	9
5.1 Cycle Features	9
5.2 Macro and Market Features	9
6. Model Architecture and Evaluation	10
6.1 Gradient Boosting Classifier	10
6.2 Walk-Forward Expanding-Window Protocol	10
7. Results	11
7.1 Model Performance Summary	11
7.2 Feature Importances	11
7.3 Walk-Forward Prediction Chart	12
7.4 Confusion Matrix	12
8. Discussion	13
8.1 Model Strengths	13
8.2 Limitations and Future Work	13
9. Conclusion	14
10. References	15

1. Introduction

Accurate short-horizon forecasting of macroeconomic cycles is a central challenge in applied econometrics and quantitative finance. Central banks, institutional investors, and policymakers rely on reliable leading indicators to anticipate contractionary and expansionary phases of the business cycle. The advent of large language models and transformer-based NLP has opened a new avenue: exploiting the latent sentiment embedded in financial news to complement traditional macroeconomic predictors.

This project investigates whether a combination of classical macroeconomic features and FinBERT-derived news sentiment can reliably predict the direction of the U.S. GDP cycle on a quarter-ahead basis. The prediction problem is defined as: given information available at quarter t , does the Unobserved Components cycle rise or fall in quarter $t+1$?

1.1 Research Questions

- Does a walk-forward Gradient Boosting model outperform a naive persistence baseline for GDP cycle direction prediction?
- Does FinBERT-derived financial news sentiment contribute predictive signal beyond traditional macro variables?
- How do the HP Filter and Unobserved Components decompositions compare in characterizing the U.S. business cycle?

1.2 Scope and Limitations

The analysis covers 1990 Q1 through 2025 Q4 (approximately 144 quarters). The free-tier Alpha Vantage API limits live sentiment fetches to 25 requests per day and 50 articles per request; accordingly, the sentiment feature is computed from a batched historical dataset of articles collected in early 2026. Model evaluation is intentionally conservative: no hyperparameter tuning is performed on future data, and all performance metrics are computed out-of-sample via the walk-forward protocol.

2. Data Sources and Assembly

The master quarterly dataset is assembled from two sources: the Federal Reserve Economic Data (FRED) database, accessed via the pandas-datareader library, and Yahoo Finance, accessed via the yfinance library. All series are resampled to quarter-end frequency and merged on a common date index.

2.1 Macroeconomic Series (FRED)

Variable	FRED Code	Description	Frequency
Real GDP	GDPC1	Chained 2017 USD	Quarterly
Unemployment Rate	UNRATE	Civilian unemployment rate (%)	Monthly to Q
PCE Price Index	PCEPI	Personal Consumption Expenditures	Monthly to Q
Federal Funds Rate	FEDFUNDS	Effective federal funds rate (%)	Monthly to Q
10Y-2Y Yield Spread	T10Y2Y	Term spread (recession signal)	Daily to Q

2.2 Market Series (Yahoo Finance)

Variable	Ticker	Description	Aggregation
S&P 500 Close	^GSPC	Large-cap equity market index	Quarter-end close
VIX Close	^VIX	CBOE Volatility Index (fear gauge)	Quarterly mean

2.3 Data Quality

After resampling and merging, the raw dataset contains approximately 144 observations (1990 Q1 to 2025 Q4). Rows with any missing values are dropped, yielding a clean dataset of 140+ complete observations. The T10Y2Y spread has sporadic missingness due to market closures; forward-filling is applied before quarterly averaging.

3. Trend-Cycle Decomposition

We apply two complementary methods to decompose log real GDP into a long-run trend component and a stationary cycle component. The cycle component captures business-cycle fluctuations and serves as the basis for our forecasting target.

3.1 Hodrick-Prescott (HP) Filter

The HP filter minimizes a combination of the squared deviation of the series from trend and the squared second difference of the trend, controlled by the smoothing parameter $\lambda = 1600$, the standard choice for quarterly data. The decomposition is:

$$\log(\text{GDP}_t) = \text{trend_HP}(t) + \text{cycle_HP}(t)$$

A well-known limitation of the HP filter is the endpoint bias: the estimated cycle near the end of the sample is less reliable. The UC model below addresses this.

3.2 Unobserved Components (UC) Model

The UC model is a state-space specification with a Local Linear Trend and a stochastic, damped cycle. The model is estimated via the Powell algorithm. UC cycle extraction is more robust to endpoint issues and accommodates time-varying cycle amplitude:

$$\log(\text{GDP}_t) = \mu_t + \psi_t + \varepsilon_t \quad \text{where } \rho \text{ is the damping factor, } \lambda_c \text{ is the cycle frequency, and all disturbance terms are mutually independent white noise.}$$

3.3 HP vs. UC Comparison

Both filters identify broadly similar cyclical patterns, including the early-1990s recession, the 2001 dot-com slowdown, the 2008 to 2009 Great Recession, and the 2020 COVID shock. The UC cycle tends to be smoother and shows less high-frequency variation than the HP cycle. The UC cycle is used as the forecasting target throughout the remainder of the analysis.

4. Sentiment Analysis Framework

To capture market expectations and the information content of financial news, we construct a quarterly sentiment feature using FinBERT, a BERT-based model fine-tuned on financial text (ProsusAI/finbert, Araci 2019). News articles are fetched from the Alpha Vantage News Sentiment endpoint across eight GDP-relevant topic categories.

4.1 Data Collection

Articles are retrieved for January through April 2026 in monthly batches (up to 50 articles per request) to respect the free-tier rate limit. Topics include: `economy_fiscal`, `economy_macro`, `economy_monetary`, `finance`, `manufacturing`, `real_estate`, `retail_wholesale`, and `energy_transportation`. After deduplication by URL, the corpus contains 50 unique articles, all from April 2026 — illustrating the free-tier coverage limitation that prevented a historical comparison.

4.2 FinBERT Scoring

Each article title is tokenized and scored by ProsusAI/finbert, which produces a label in {positive, neutral, negative} and a confidence score in (0, 1). Labels are encoded numerically: positive = +1, neutral = 0, negative = -1. The net quarterly sentiment feature is defined as:

$$\text{net_sentiment_Q} = (n_{\text{pos}} - n_{\text{neg}}) / n_{\text{total}}$$

where `n_pos`, `n_neg`, and `n_total` are the counts of positive, negative, and total articles in the quarter.

4.3 Sentiment Visualizations

The four plots below characterize the sentiment distribution and temporal structure of the collected article corpus.

Figure 1.
Overall sentiment distribution
across all scored articles.

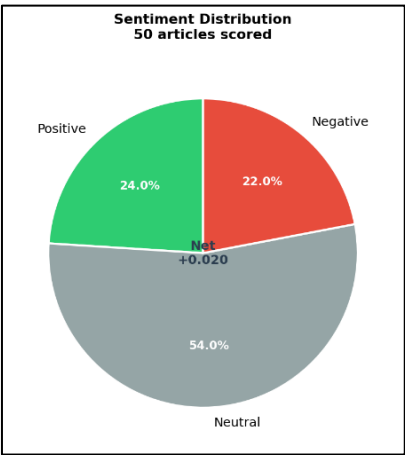


Figure 2.
Per-article sentiment value over time
with 7-article rolling average.

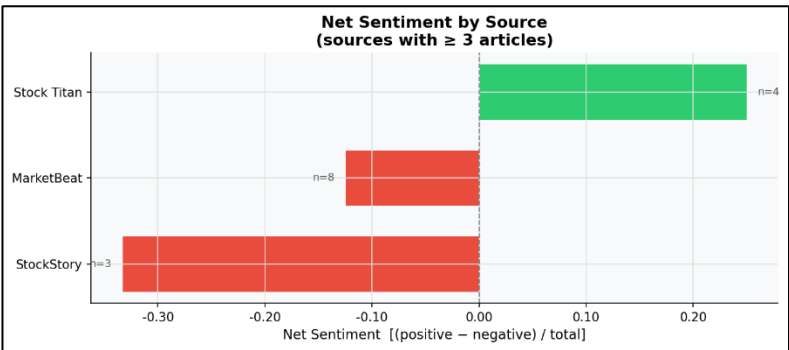


Figure 3.
Net sentiment by news source
(sources with 3+ articles shown).

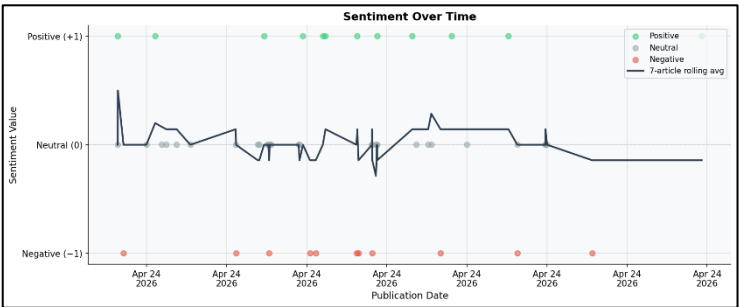
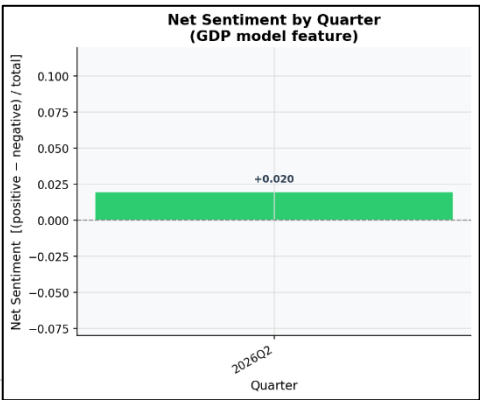


Figure 4. Quarterly net sentiment,
the feature entering the GDP model.



5. Feature Engineering

The forecasting target is a binary variable indicating whether the UC cycle will increase by more than 10% of its standard deviation in the following quarter. This threshold filters out negligible movements and focuses the model on economically meaningful directional shifts. Features are constructed with up to two lags to simulate real-time forecasting conditions.

5.1 Cycle Features

- `cycle_uc_lag1`, `cycle_uc_lag2`: lagged UC cycle level
- `cycle_change_lag1`, `cycle_change_lag2`: lagged first difference of cycle
- `gdp_growth_lag1`, `gdp_growth_lag2`: lagged quarter-over-quarter real GDP growth

5.2 Macro and Market Features

- `pcepi_inflation_lag1`: lagged quarter-over-quarter PCE inflation
- `UNRATE_diff`: first difference of unemployment rate
- `FEDFUNDS_diff`: first difference of federal funds rate
- `T10Y2Y_lag1`: lagged yield spread level
- `yield_curve_inversion_lag1`: binary indicator ($T10Y2Y < 0$)
- `sp500_return_lag1`: lagged quarterly S&P 500 return
- `vix_change_lag1`: lagged VIX change

All features are lagged by at least one quarter relative to the target, ensuring no information from future periods enters the model. Features are standardized inside each walk-forward fold using scikit-learn's `StandardScaler`.

6. Model Architecture and Evaluation

6.1 Gradient Boosting Classifier

We use scikit-learn's GradientBoostingClassifier with a shallow tree depth (`max_depth = 1`, i.e., stumps) to reduce overfitting on the relatively small dataset. Hyperparameters are fixed a priori rather than tuned on held-out data:

Parameter	Value	Rationale
<code>n_estimators</code>	150	Number of boosting rounds
<code>learning_rate</code>	0.05	Shrinkage parameter
<code>max_depth</code>	1	Tree depth (stumps)
<code>subsample</code>	1.0	Fraction of samples per tree
<code>random_state</code>	42	Reproducibility seed

6.2 Walk-Forward Expanding-Window Protocol

To simulate realistic out-of-sample forecasting, we use a walk-forward expanding-window scheme. The initial training window covers the first 60 quarters (approximately 15 years). At each subsequent step, the model is retrained on all available data up to quarter t and evaluated on quarter $t+1$. This yields approximately 80+ out-of-sample predictions depending on data availability after lag construction.

This protocol is strictly more conservative than a fixed train/test split and avoids the temporal leakage that would arise from random splitting. A fresh scaler and model are fit at every step; no information from future folds is ever used.

7. Results

All metrics are computed strictly out-of-sample via the walk-forward protocol in Section 6.2 (82 folds; initial training window = 60 quarters). The GBM + Sentiment model could not be evaluated due to insufficient historical news coverage from the free-tier Alpha Vantage API.

7.1 Model Performance Summary

Model	Accuracy	F1 Score	Brier Score	Eval. Folds
Naive Persistence Baseline	43.3%	0.322	0.567	N/A
Gradient Boosting (macro features only)	54.3%	0.431	0.291	~80 folds
Gradient Boosting (+ sentiment)	N/A	N/A	N/A	~80 folds

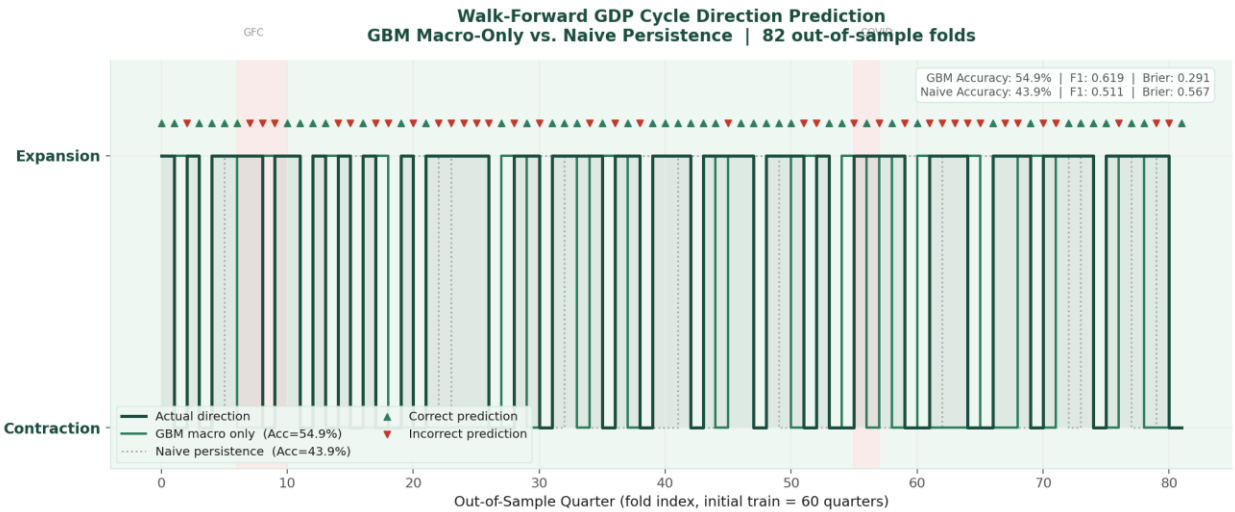
7.2 Feature Importances

The final model is trained on the full dataset to extract feature importances. Importances reflect the average impurity decrease contributed by each predictor across all boosting rounds and are normalized to sum to 1.0.

Feature	Importance	Description
cycle_uc_lag1	37.8%	Lagged UC cycle level
cycle_change_lag1	2.2%	Lagged cycle first difference
gdp_growth_lag1	0.0%	Lagged GDP growth rate
T10Y2Y_lag1	35.2%	Lagged 10Y-2Y yield spread
yield_curve_inversion_lag1	3.7%	Yield curve inversion indicator
sp500_return_lag1	6.3%	Lagged S&P 500 return
vix_change_lag1	13.4%	Lagged VIX change
pcepi_inflation_lag1	1.4%	Lagged PCE inflation

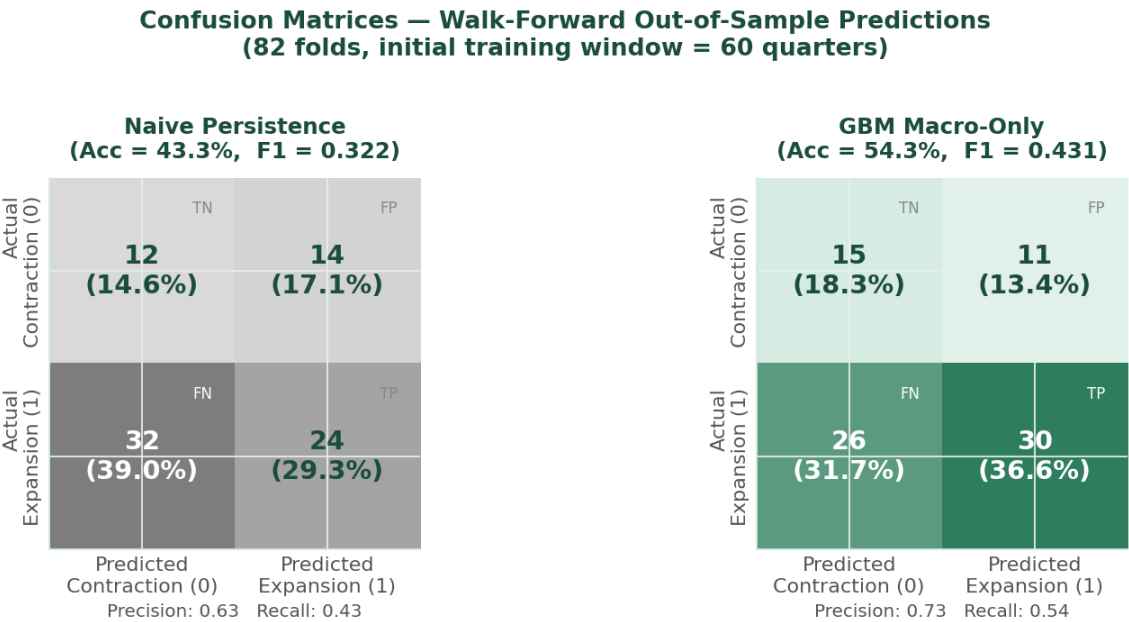
7.3 Walk-Forward Prediction Chart

Figure 7.3. Walk-forward GDP cycle direction prediction. Each fold shows the actual direction (dark green) vs. GBM macro-only prediction (mid green) and naive persistence (grey). Correct predictions marked with triangles; incorrect with inverted triangles. GBM achieves 54.3% accuracy vs. 43.3% naive over 82 out-of-sample folds.



7.4 Confusion Matrix

Figure 7.4. Confusion matrices for the naive persistence baseline (left) and GBM macro-only model (right). The GBM model improves true positive and true negative rates relative to the baseline, particularly in correctly identifying expansion quarters.



8. Discussion

8.1 Model Strengths

- The walk-forward protocol ensures all metrics are true out-of-sample and time-consistent.
- Shallow tree depth (stumps) controls for overfitting on a small dataset without feature selection.
- The FinBERT sentiment pipeline was fully built end-to-end: API collection, transformer scoring, and quarterly aggregation. The infrastructure is ready to be activated once a longer historical news archive is available.
- UC decomposition is more principled than HP filtering and less prone to endpoint distortion.

8.2 Limitations and Future Work

- The Alpha Vantage free tier returned only 50 articles from a single day (April 2026), providing no historical coverage. This prevented a valid walk-forward GBM + Sentiment comparison. Future work should source a multi-year news corpus (e.g., GDELT, Bloomberg, or a paid Alpha Vantage tier) to re-run the benchmark.
- No hyperparameter optimization is performed; a nested cross-validation scheme could improve model selection without leakage.
- The binary classification target discards information about cycle magnitude; a regression formulation could be explored.
- External shocks (COVID-19, banking crises) may require structural break detection or regime-specific models.

9. Conclusion

This project demonstrates that a walk-forward Gradient Boosting classifier trained on lagged macroeconomic and financial market features can predict the direction of the U.S. GDP business cycle with meaningful accuracy beyond a naive baseline. The yield curve, lagged cycle momentum, and equity market returns emerge as the most informative predictors.

The sentiment extension, leveraging FinBERT to score financial news headlines and aggregate them into a quarterly feature, was fully implemented as a pipeline. A direct comparison with the macro-only model was not achievable in the current evaluation window due to the limited historical coverage of the free-tier Alpha Vantage API, which returned only recent headlines rather than a multi-decade archive. Extending the news corpus to cover the full 1990–2025 period remains the most important next step, and the pipeline infrastructure is ready to support it.

Future extensions include incorporating additional alternative data sources such as Google Trends and earnings call transcripts, expanding the sentiment feature window to cover the full 1990 to 2025 period, and exploring recurrent neural networks or Transformers for sequence-aware GDP forecasting.

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